

Ricardo Barrios

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EDUCATION

University of Magdalena

Bachelor of Science in Systems Engineering

Expected July 2026

- Relevant Coursework: Artificial Intelligence, Web Development, Operating Systems, UI Design, Software Architecture.
- Achievements: 4.2 CGPA

PROJECTS

Blind-First Navigation App | *React Native, Node.js, MongoDB, mobile development*

March 2025 - Present

- Led a team of four to build a voice-controlled navigation app for visually impaired users using React Native & Node.js.
- Implemented a speech recognition system with over 90% accuracy and optimized graph-based routing with Dijkstra's algorithm for fast, hands-free navigation.
- Achieved route recalculation in under 3 seconds and optimized database queries, reducing delays by 40% using a NoSQL database structure.

Corporate Website for Engineering Firm | *TypeScript, Next.js, React, Tailwind, Netlify, CMS*

February 2025 – March 2025

- Built and deployed INGENOVA's first corporate website to establish a digital presence in the civil engineering sector.
- Utilized a modern tech stack to streamline content management and facilitate rapid updates by non-technical staff, reducing reliance on external developers.
- Enabled content changes in under 2 minutes and secured exceptional performance metrics (Google Lighthouse scores: 99 Performance, 96 Accessibility, 96 Best Practices, 100 SEO).

AI-Powered Mango Classification | *Python, TensorFlow, scikit-learn, deep learning*

October 2024 – November 2024

- Developed an AI system to classify mango ripeness using a dataset of 2,714 images.
- Spearheaded a team of four, optimizing dataset preprocessing and incorporating image augmentation techniques across multiple models (Deep and Shallow Neural Networks, Random Forest, Decision Trees, Linear Regression).
- Increased model accuracy from 70% to over 95%, significantly enhancing decision-making in agricultural supply chains and reducing waste.

Habit Tracker | *React, JavaScript, CSS, IndexedDB, Local Storage, PWA*

August 2024 – October 2024

- Built a habit tracking application with offline functionality and gamified elements for personal use.
- Designed a React-based progressive web app with an offline-first architecture and integrated persistent local storage.
- Enhanced productivity by ensuring data retention across device restarts while boosting engagement through gamification.

Secure Electronic Voting System | *PHP, JavaScript, Material Design, MySQL, authentication*

March 2024 – May 2024

- Led a two-student team in developing a secure electronic voting system to modernize traditional election processes.
- Engineered both frontend and backend components, incorporating user authentication, email notifications, database interactions, and role-based access control to ensure election security.
- Ensured 100% voter privacy and system integrity, providing a streamlined and user-friendly election process, successfully managing a test with thirty simultaneous voters without downtime or security breaches.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C/C++, HTML/CSS, PHP

Frameworks: React, Next.js, Tailwind, Node.js, Django, Spring Boot

Developer Tools: Git, Docker, CI/CD, Power BI, Google Cloud, TensorFlow, Pandas, scikit-learn

Databases: PostgreSQL, MySQL, SQLite, MongoDB

Spoken Languages: English (Fluent), Spanish (Fluent), French (A2-B1)